

Claude Code CLI — EntraID JWT Authentication via AIGW

Version 2. This guide supersedes the previous revision of this document. If you configured the app registration against an earlier version of this guide, work through the setup steps below in full before retrying — the claims configuration surface has changed and the two approaches should not be mixed.

Configures the Claude Code CLI to send requests through this self-hosted AIGW gateway (although this process works with the SaaS AIGW offering too), authenticated using a Microsoft EntraID JWT bearer token obtained automatically from the local Azure CLI session. User and workspace assignment is driven entirely by **Entra security groups**, so adding or moving a user is a group-membership change — no per-user policy edits.

How it works

Claude Code CLI supports an `apiKeyHelper` — a script it calls on each session start whose stdout becomes the bearer token sent to the gateway. `get-az-token.sh` in this repo fills that role: it fetches an EntraID access token from the local Azure CLI cache, triggering an interactive browser login if the session has expired.

```
Claude Code CLI
├── calls get-az-token.sh → outputs EntraID access token
└── sends requests to AIGW gateway (http://localhost:8787)
    └── Authorization: Bearer <token>
        └── gateway validates JWT, maps claims → Portkey workspace
```

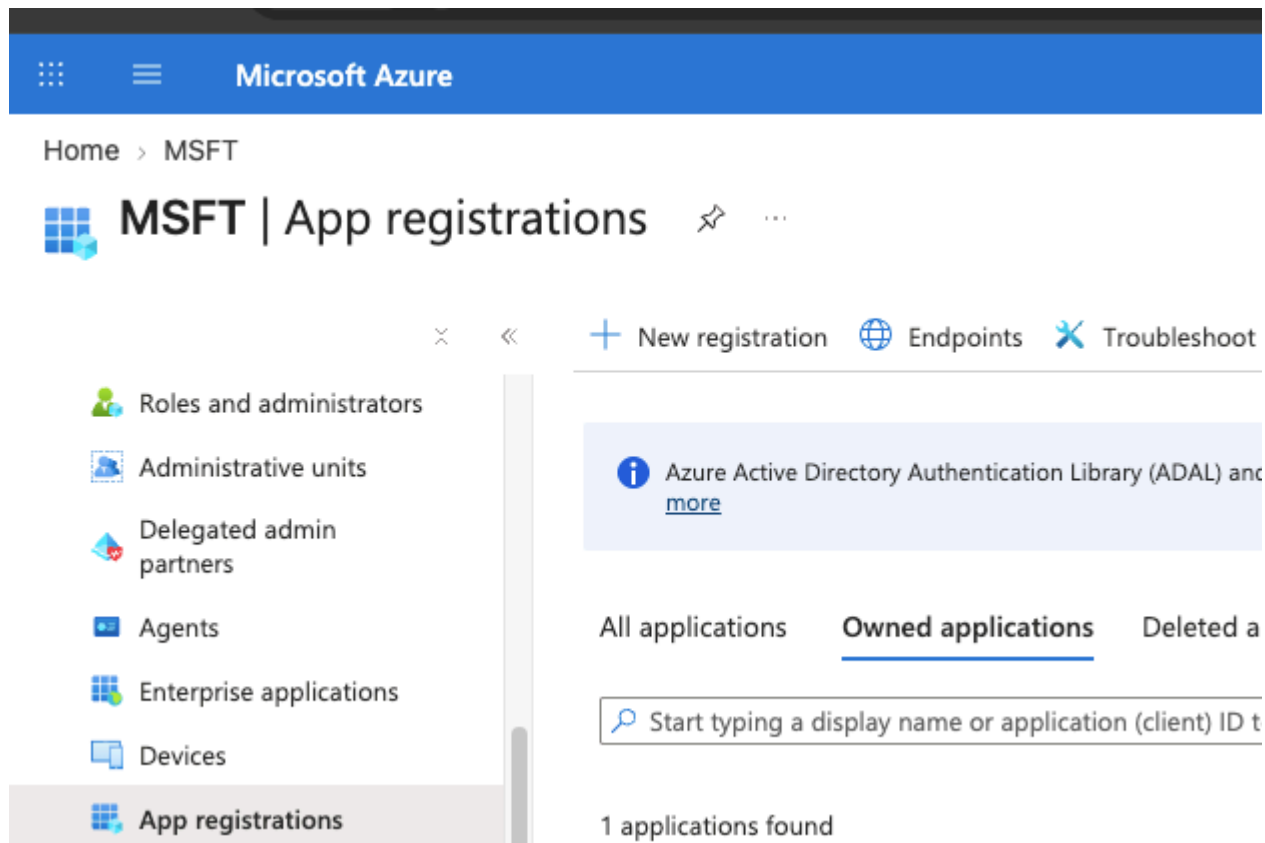
Unlike an interactive OIDC flow, the Azure CLI requests an **access token for a resource**, not an ID token — which means the app registration must additionally expose an API and pre-authorize the Azure CLI as a client (steps 6–7 below).

Prerequisites

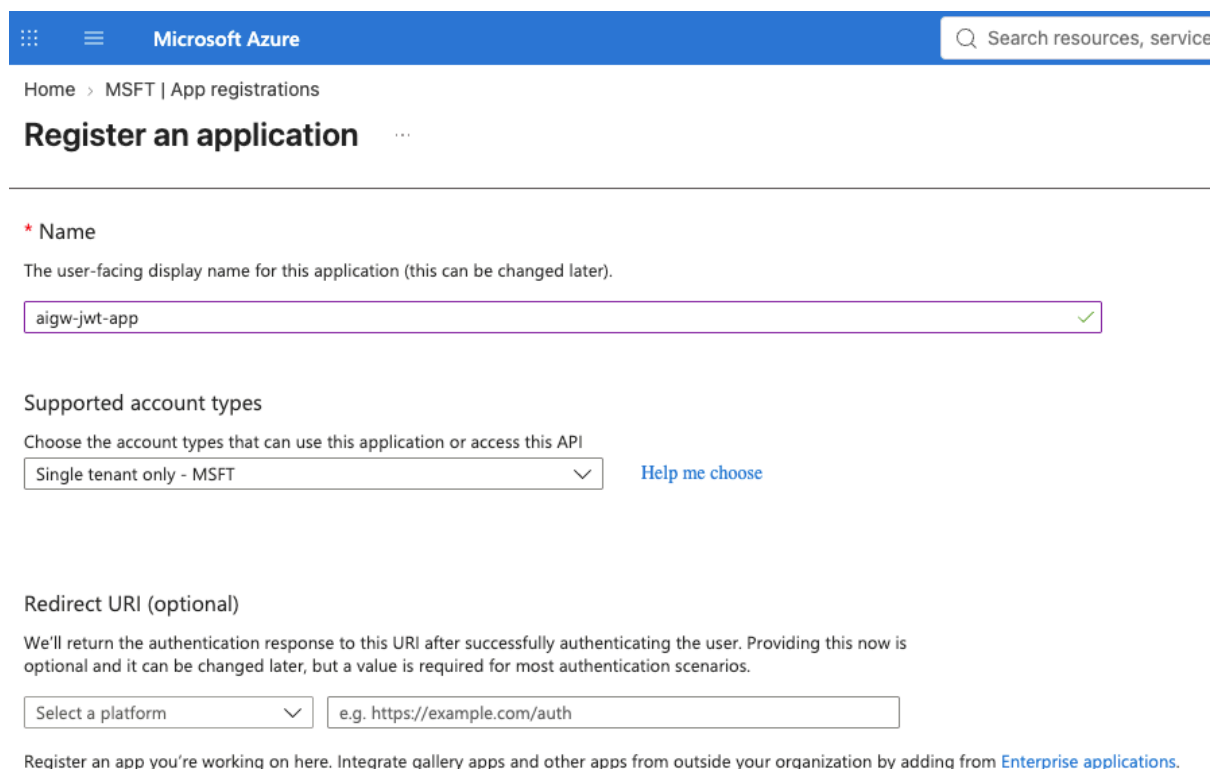
- This repo's `docker-compose.yml` stack running
 - `JWT_ENABLED: ON`
- An EntraID tenant with permissions to register applications and manage Enterprise Application properties (Application Administrator or Cloud Application Administrator)
- `openssl` installed locally (for generating the token-signing certificate in step 2)
- Azure CLI installed (`brew install azure-cli`) and logged in at least once

1. Register the application

In the [Entra Admin Center](#), go to **App registrations** → **New registration**.



- **Name:** something identifiable, e.g. `aigw-jwt-app`
- **Supported account types:** Single tenant
- **Redirect URI:** not required for the CLI token flow — leave blank unless you also plan to use this app registration for an interactive OIDC client



Note the **Application (client) ID** and **Directory (tenant) ID** from the app's Overview page — both are needed throughout this guide.

2. Generate and upload a token-signing certificate

Customizing claims on this app (step 5) requires a certificate — Entra signs tokens with a custom key once you add custom claims, instead of the default Microsoft key. Generate a self-signed certificate and upload the public half:

```
openssl req -x509 -newkey rsa:2048 \  
-keyout key.pem -out cert.pem \  
-days 365 -nodes \  
-subj "/CN=PortkeyTokenSigningKey"
```

In **App registrations** → **your app** → **Certificates & secrets** → **Certificates** → **Upload certificate**, upload **cert.pem** (the public key only — never upload or share **key.pem**).

Upload certificate

✓ Upload Completed for cert.pem
1.12 KiB | "Streaming upload"

Upload a certificate (public key) with one of the following file types: .cer, .pem, .cert *

cert.pem

Description

Enter a description for this certificate

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo and a search bar. The breadcrumb trail indicates the path: Home > MSFT | App registrations > aigw-jwt-app. The main heading is 'aigw-jwt-app | Certificates & secrets'. On the left, a sidebar lists various management options: Overview, Quickstart, Integration assistant, Diagnose and solve problems, Manage (expanded), Branding & properties, Authentication (Preview), Certificates & secrets (selected), Token configuration, API permissions, Expose an API, App roles, Owners, Roles and administrators, and Manifest. The main content area features a feedback prompt, a descriptive paragraph about credentials, and a tabbed interface for 'Certificates (1)', 'Client secrets (0)', and 'Federated credentials (0)'. The 'Certificates' tab is active, showing a table with one certificate entry. The table has columns for Thumbprint, Description, Start date, and Expires. The entry has a thumbprint of DA3491AE646C99900F2BA45AB9B3F..., a description of CN=PortkeyTokenSigningKey, a start date of 10/2/2026, and an expiration date of 10/1/2028. An 'Upload certificate' button is located above the table.

Thumbprint	Description	Start date	Expires
DA3491AE646C99900F2BA45AB9B3F...	CN=PortkeyTokenSigningKey	10/2/2026	10/1/2028

Keep `key.pem` safe — treat it like any other private key material. It is not consumed by the gateway or by Claude; it only needs to exist so the certificate it backs is valid.

3. Patch the manifest

In the app's **Manifest** tab, set two fields under `"api"`:

```
"api": {
  "acceptMappedClaims": true,
  "requestedAccessTokenVersion": 2,
  ...
}
```

Microsoft Azure

Q Si

Home > MSFT | App registrations > aigw-jwt-app

aigw-jwt-app | Manifest

Search

Refresh

Got feedback?

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Branding & properties

Authentication (Preview)

Certificates & secrets

Token configuration

API permissions

Expose an API

App roles

Owners

Roles and administrators

Manifest

An app manifest is a JSON representation of an app registration. The Microsoft system will automatically update the other one for them to match.

Microsoft Graph App Manifest (New)

AAD Graph App Manifest (Depr

See [Understanding the Microsoft Graph application manifest](#) to learn how to ec

Save

Discard

Download

Upload

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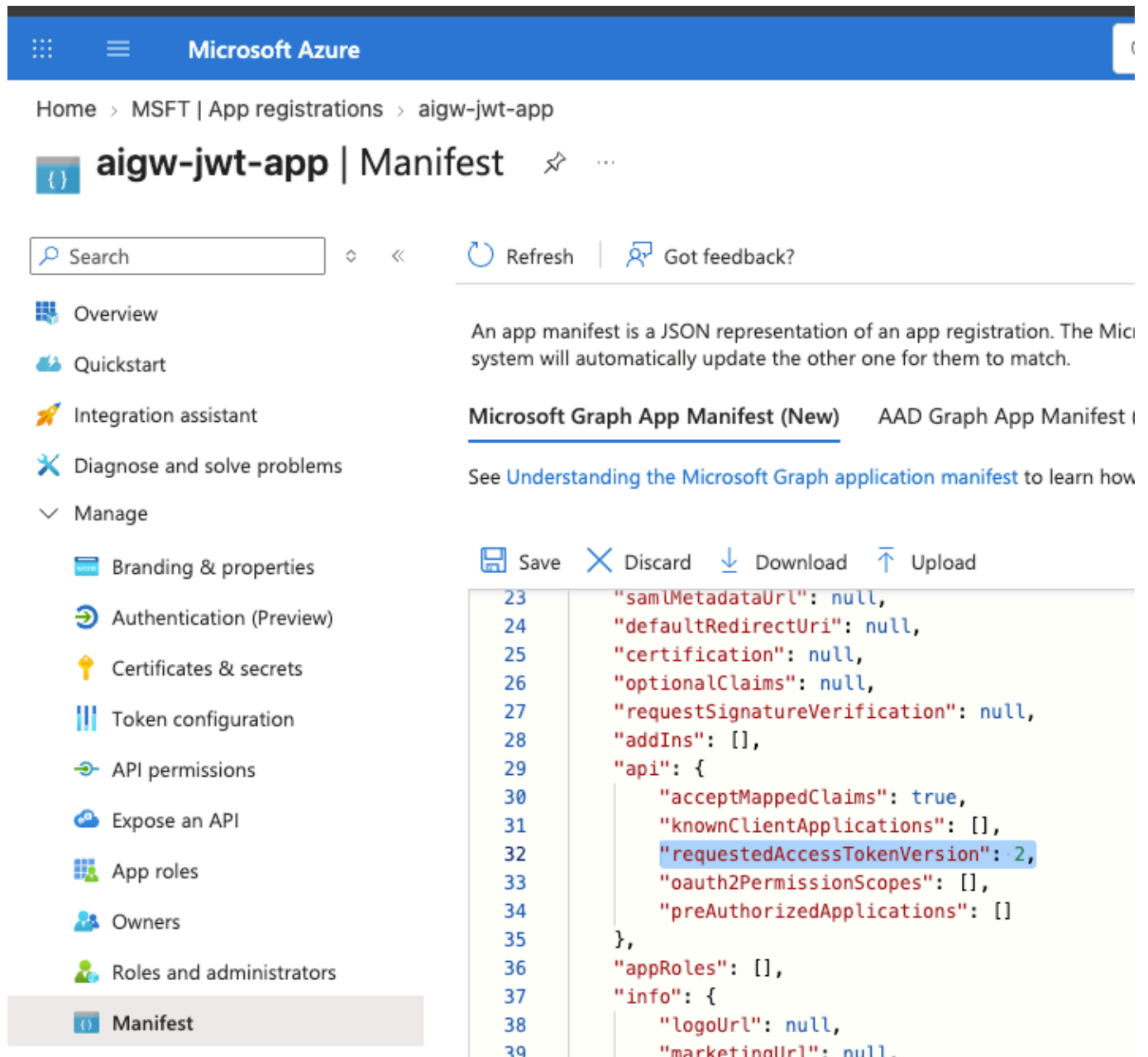
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Home > MSFT | App registrations > aigw-jwt-app

aigw-jwt-app | Manifest

Search Refresh Got feedback?

Overview

An app manifest is a JSON representation of an app registration. The Microsoft system will automatically update the other one for them to match.

Microsoft Graph App Manifest (New) AAD Graph App Manifest

See [Understanding the Microsoft Graph application manifest](#) to learn how

Save Discard Download Upload

```

23 "samlMetadataUrl": null,
24 "defaultRedirectUri": null,
25 "certification": null,
26 "optionalClaims": null,
27 "requestSignatureVerification": null,
28 "addIns": [],
29 "api": {
30   "acceptMappedClaims": true,
31   "knownClientApplications": [],
32   "requestedAccessTokenVersion": 2,
33   "oauth2PermissionScopes": [],
34   "preAuthorizedApplications": []
35 },
36 "appRoles": [],
37 "info": {
38   "logoUrl": null,
39   "marketingUrl": null,

```

- `requestedAccessTokenVersion: 2` — without this, EntraID issues v1.0 tokens, which `acceptMappedClaims` and the audience checks below do not expect.
- `acceptMappedClaims: true` — required for the app to emit the custom claims configured in step 5.

Click **Save**.

4. Restrict and assign access via the Enterprise Application

In the [Entra Admin Center](#), go to **Enterprise applications** → **All applications** and search for your app by name — the App Registration above automatically has a matching Enterprise Application (service principal).

Home > Enterprise applications

Enterprise applications | All applications

MSFT

<

<<

+ New application

↻ Refresh

⬇ Download (Export)

ℹ Preview info

≡ Columns

🔍 Preview features

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Overview

Manage

All applications

Private Network connectors

User settings

App launchers

Custom authentication extensions

View, filter, and search applications in your organization that are set up to use your Microsoft Entra tenant as their Identity Provider.

The list of applications that are maintained by your organization are in [application registrations](#).

aigw

✕

Application type == Enterprise Applications ✕

Apply

1 application found

Name	↕	Object ID	Application ID	Homepage URL	Created on
A aigw-jwt-app		4c8ff76a-c68a-42a9-8dd3-...	c5490456-8e83-4df6-a979-...		10/2/2026

4a. Require assignment

Under **Properties**, set **Assignment required?** to **Yes**. This is what makes group membership meaningful – without it, any user in the tenant can sign in and get a token regardless of group assignment below.

Home > Enterprise applications | All applications > aigw-jwt-app

aigw-jwt-app | Properties

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

Security

Activity

Help

View and manage application settings for your organization. Editing properties like display information, user sign-in settings, and user visibility settings requires Global Administrator, Cloud Application Administrator, Application Administrator roles. [Learn more](#).

If this application resides in your tenant, you can manage additional properties on the [application registration](#).

Activation status ⓘ

Enabled for users to sign-in? ⓘ

Name * ⓘ

Homepage URL ⓘ

Logo ⓘ

Application ID ⓘ

Object ID ⓘ

Assignment required? ⓘ

Visible to users? ⓘ

Activated

Yes

No

aigw-jwt-app ✓

A

Select a file

📁

c5490456-8e83-4df6-a979-9ed2b4dd84a5

4c8ff76a-c68a-42a9-8dd3-7832df6b03a4

Yes

No

Yes

No

4b. Assign the groups that should have access

Under **Users and groups** → **Add user/group**, assign the security group(s) whose members should be able to use this gateway (e.g. **AIRS Group**). Only members of an assigned group can sign in once assignment is required.

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Home > Enterprise applications | All applications > aigw-jwt-app

aigw-jwt-app | Users and groups

Enterprise Application

◊ << + Add user/group ✎ Edit assignment 🗑️ Remo

- Overview
- Deployment Plan
- Diagnose and solve problems
- Manage
 - Properties
 - Owners
 - Roles and administrators
 - Users and groups**

The application will not appear for assigned users with

Assign users and groups to app-roles for your application

First 200 shown, search all users & groups

Display name

No application assignments found

Home > Enterprise applications | All applications > aigw-jwt-app | Users and groups

Add Assignment

MSFT

Users and groups

None Selected

Select a role

Default Access

Users and groups

Try changing or adding filters if you don't see what you're looking for.

Search

AIRS Group

1 result found

All Users Agents' user accounts Groups

	Name	Type	Details
☒	AG AIRS Group	Group	

Onboarding a new user: add them to the appropriate Entra group. No app registration change, no policy edit, no restart — their next token pulls access and claims automatically. This is the mechanism for allocating users "on the fly."

5. Configure claims

Still on the Enterprise Application, go to **Single sign-on** → **OIDC-based Sign-on** → **Attributes & Claims** → **Edit**.

Home > Enterprise applications | All applications > aigw-jwt-app

aigw-jwt-app | OIDC-based Sign-on

Enterprise Application

Overview

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Diagnose and solve problems

Manage

- Properties
- Owners
- Roles and administrators
- Users and groups
- Single sign-on
- Provisioning
- Application proxy
- Self-service
- Custom security attributes

OIDC Applications require custom signing keys to customize claims. Please check the security considerations before customizing the claims for the application. [Learn more](#)

This application uses OpenID Connect and OAuth. This protocol simplifies application configuration, has easy-to-use SDKs, and enables your application to use MS Graph. [Learn more](#)

Because this application uses OpenId Connect and OAuth, most single sign-on configuration is already complete. [Learn more about application and service principal objects in Microsoft Entra](#)

1

Configure application properties

[Go to application](#)

Please go to aigw-jwt-app in the App registrations experience to edit properties such as reply Urls, identifiers, optional claims, among others. Your account should have the required permissions (Global Administrator, Cloud Application Administrator, Application Administrator, or owner of the application object). [Learn more about admin roles in Microsoft Entra](#)

2

Attributes & Claims

[Edit](#)

This application does not have any JWT-based claims configured. Click on edit to start configuring claims.

Add the claims the gateway needs as **Additional claims** (use **+ Add new claim**, source = typed constant value, not an attribute):

Claim name	Value	Purpose
portkey_oid	Your deployment's ORGANISATIONS_TO_SYNC UUID	Org routing — the gateway rejects tokens missing this
portkey_workspace	A workspace slug, e.g. ws- main-a-997260	Workspace routing
scopes	e.g. completions.write	Overrides JWT_LOCAL_AUTH_DEFAULT_SCOPES explicitly

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Attributes & Claims

...

+ Add new claim

+ Add a group claim

≡ Columns

|  Got feedback?

 OIDC Applications require custom signing keys to customize claims. Please check the security considerations before customizing the claims for the application. [Learn more.](#)

Required claim

Claim name	Type	Value
No claims configured		

Additional claims

Claim name	Type	Value
portkey_oid	JWT	"95cb67bf-6fb5-4581-ac... ...
portkey_workspace	JWT	"ws-main-a-997260" ...
scopes	JWT	"completions.write" ...

NOTE: You can choose to leave out `portkey_workspace` and leverage SCIM group assignment to allocate the workspace instead. Upon authentication the gateway will scan for valid users based on the UPN or email and assign them to the appropriate workspace.

Group-based claim values (multi-team / multi-workspace)

If different teams need different values — different workspaces, or extra metadata — don't hardcode a single value. Instead, add **claim conditions** scoped to a group, with a distinct value per group. For example, a `metadata` claim that varies by team:

Manage claim

...

 Save

 Discard changes

|  Got feedback?

Name *

metadata

Namespace

Enter a namespace URI

Choose name format

Source

☒ Attribute

☐ Transformation

☐ Directory schema extension

Source attribute

Select from drop down or type a constant

Claim conditions

Returns the claim only if all the conditions below are met.

 Multiple conditions can be applied to a claim. When adding conditions, order of operation is important. [Read the documentation](#) for more information.

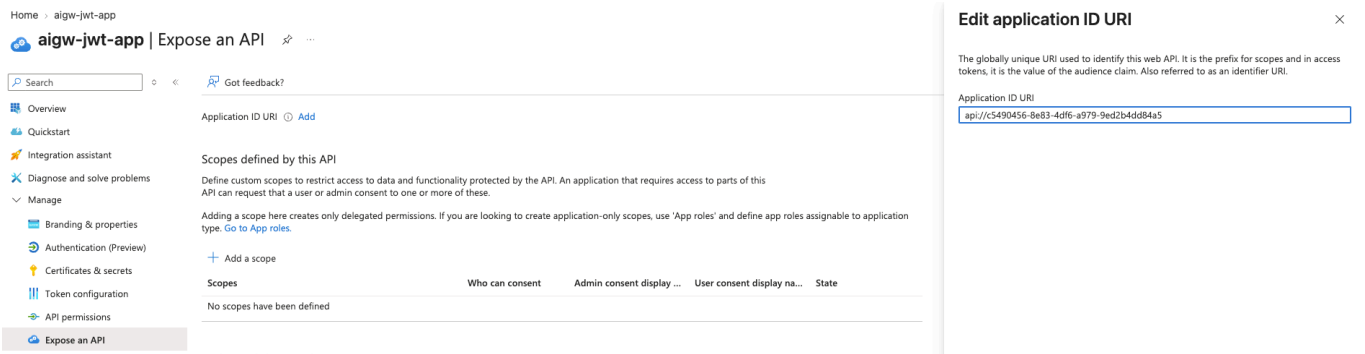
User type	Scoped Groups	Source	Value
Members	1 groups	Attribute	"{ \"environment\": \"production\", \"team\": \"backend\" }"
Members	1 groups	Attribute	"{ \"environment\": \"production\", \"team\": \"frontend\" }"
Select from drop down	Select groups	Select claim condition source.	
Advanced SAML claims options			

With this in place, the same claim resolves differently depending on which group signs in — this is how `portkey_workspace` itself can also be made group-dependent for a multi-workspace deployment: add one claim condition row per group, each with its own value, instead of a single static value in the table above.

Click **Save**.

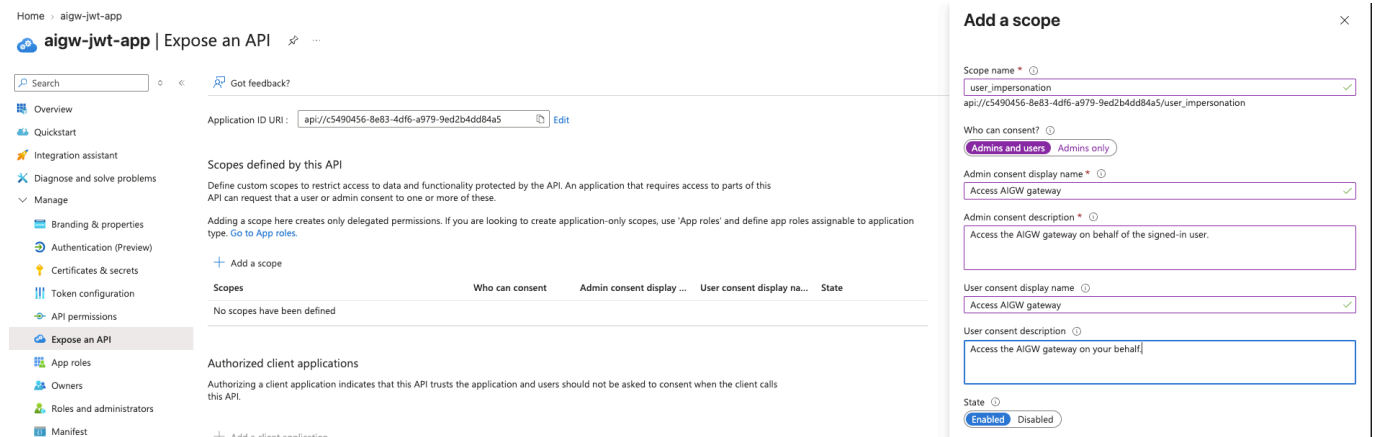
6. Expose an API

The Azure CLI requests a token *for* this app as a resource, which requires the app to expose an API. In **App registrations** → **your app** → **Expose an API**, set the **Application ID URI** to the default suggested value (`api://<client-id>`):



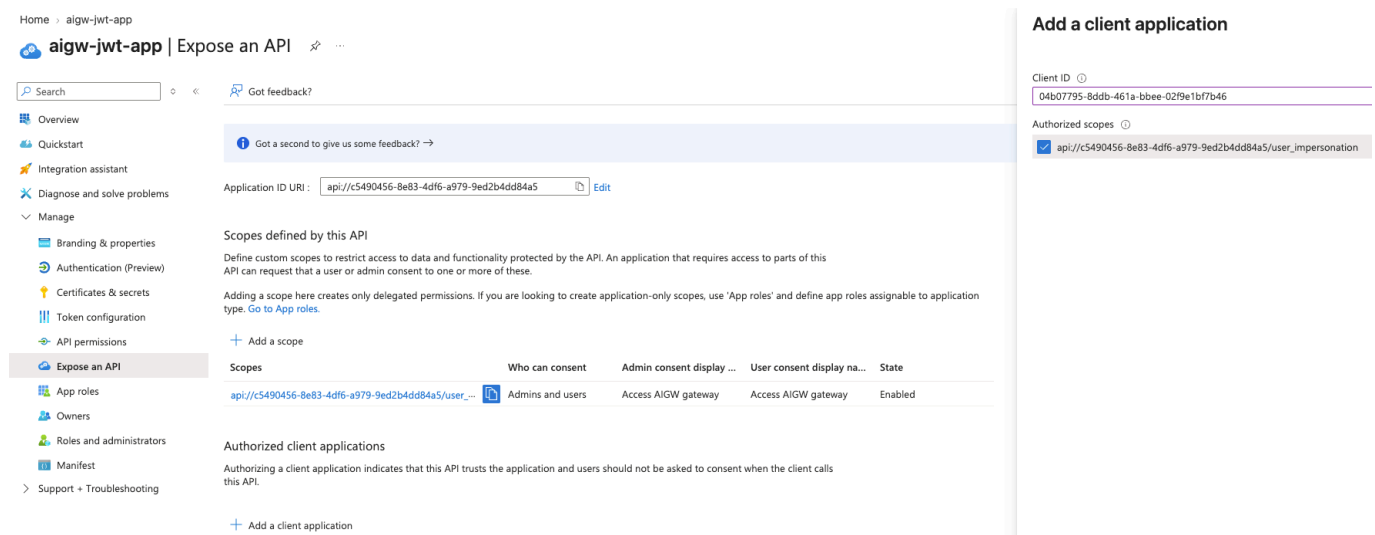
Add a scope:

Field	Value
Scope name	<code>user_impersonation</code>
Who can consent	Admins and users
Admin consent display name	<code>Access AIGW gateway</code>
Admin consent description	<code>Access the AIGW gateway on behalf of the signed-in user.</code>
User consent display name	<code>Access AIGW gateway</code>
User consent description	<code>Access the AIGW gateway on your behalf.</code>
State	Enabled



7. Pre-authorize the Azure CLI

Still on **Expose an API**, under **Authorized client applications** → **Add a client application**, add the Azure CLI's well-known client ID (`04b07795-8ddb-461a-bbee-02f9e1bf7b46`) and check the `user_impersonation` scope:



Without this step, `az account get-access-token/az login --scope` fails with `AADSTS650057` (if the API isn't exposed yet) or `AADSTS65001` (if the Azure CLI isn't pre-authorized).

Note on audience: `acceptMappedClaims` requires the token audience to be the bare application GUID. Always use the client ID (e.g. `c5490456-...`) as the resource when calling `az account get-access-token` — not the `api://` URI form. Using the URI form triggers `AADSTS501461`.

8. Get the Application (client) ID

From either the App Registration's Overview or the Enterprise Application's Overview — they share the same **Application ID**.

Home > Enterprise applications | All applications



aigw-jwt-app | Overview

Enterprise Application

Overview

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Properties

A

Name ⓘ

aigw-jwt-app

Application ID ⓘ

c5490456-8e83-4df6-a9...

Object ID ⓘ

4c8ff76a-c68a-42a9-8dd...

Copied

Getting Started

9. Install the token helper script

Copy `get-az-token.sh` from this repo to your Claude config directory:

```
cp get-az-token.sh ~/.claude/get-az-token.sh
chmod +x ~/.claude/get-az-token.sh
```

The script reads `ENTRAID_CLIENT_ID` and `ENTRAID_TENANT_ID` from the environment. Supply them via the `env` block in Claude Code's settings (step 10) — that is the recommended approach and no other configuration file is needed.

10. Configure Claude Code CLI

Edit (or create) `~/.claude/settings.json`:

```
{
  "apiKeyHelper": "/Users/<you>/.claude/get-az-token.sh",
  "env": {
    "ANTHROPIC_BASE_URL": "http://localhost:8787",
    "ENTRAID_CLIENT_ID": "<your-app-client-id>",
    "ENTRAID_TENANT_ID": "<your-tenant-id>",
    "ANTHROPIC_DEFAULT_SONNET_MODEL": "anthropic.claude-sonnet-5",
    "ANTHROPIC_DEFAULT_OPUS_MODEL": "anthropic.claude-opus-4-8",
    "ANTHROPIC_DEFAULT_HAIKU_MODEL": "anthropic.claude-haiku-4-5-20251001",
    "CLAUDE_CODE_API_KEY_HELPER_TTL_MS": "3000000"
  }
}
```

Field	Value
<code>apiKeyHelper</code>	Absolute path to <code>get-az-token.sh</code>
<code>ANTHROPIC_BASE_URL</code>	Your gateway URL — use <code>http://localhost:8787</code> for local Docker
<code>ENTRAID_CLIENT_ID</code>	The Application (client) ID from the EntraID app registration
<code>ENTRAID_TENANT_ID</code>	Your Entra tenant ID
<code>ENTRAID_RESOURCE_URI</code>	(optional) Resource to request a token for — defaults to the bare <code>ENTRAID_CLIENT_ID</code> GUID. Override only if you have a reason to deviate; do not set it to the <code>api://</code> URI form (see the audience note above).
<code>ANTHROPIC_DEFAULT_SONNET_MODEL</code>	Model ID Claude Code uses for Sonnet requests — must include the <code>anthropic.</code> prefix when routing via Vertex AI
<code>ANTHROPIC_DEFAULT_OPUS_MODEL</code>	Model ID for Opus requests — same prefix requirement
<code>ANTHROPIC_DEFAULT_HAIKU_MODEL</code>	Model ID for Haiku requests — same prefix requirement
<code>CLAUDE_CODE_API_KEY_HELPER_TTL_MS</code>	How long (ms) Claude Code caches the token from <code>apiKeyHelper</code> — set to <code>3000000</code> (~50 min) to match the Azure CLI token lifetime and avoid unnecessary re-fetches

Tip: If you have `ENTRAID_CLIENT_ID` and `ENTRAID_TENANT_ID` in your shell environment already (e.g. via `direnv`), you can omit those two from the `env` block and the script will pick them up directly. The model vars should always be set explicitly when using this config.

Alternatively if you do not want to use the script, you can call `az account get-access-token --resource <ENTRAID_CLIENT_ID> --tenant <ENTRAID_TENANT_ID> --query "accessToken" -o tsv` as the `apiKeyHelper` directly

Project-level config (recommended when mixing auth methods)

If you use a different Claude Code config for other repos (e.g. a direct Portkey cloud setup), put the gateway settings in a **project-level** `.claude/settings.json` at the root of this repo instead of the global `~/.claude/settings.json`. Claude Code merges the two — project settings override global ones for the same keys — so your global config stays intact for everything else:

```
mkdir -p .claude
# create .claude/settings.json with only the gateway-specific overrides
```

```
{
  "apiKeyHelper": "/Users/<you>/.claude/get-az-token.sh",
```

```

"env": {
  "ANTHROPIC_BASE_URL": "http://localhost:8787",
  "ENTRAID_CLIENT_ID": "<your-app-client-id>",
  "ENTRAID_TENANT_ID": "<your-tenant-id>",
  "ANTHROPIC_CUSTOM_HEADERS": "x-portkey-config: <your-config-id>",
  "ANTHROPIC_DEFAULT_SONNET_MODEL": "anthropic.claude-sonnet-5",
  "ANTHROPIC_DEFAULT_OPUS_MODEL": "anthropic.claude-opus-4-8",
  "ANTHROPIC_DEFAULT_HAIKU_MODEL": "anthropic.claude-haiku-4-5-20251001",
  "CLAUDE_CODE_API_KEY_HELPER_TTL_MS": "3000000"
}
}

```

Theme, permissions, and any other global settings are inherited automatically.

11. Set the Portkey config header (optional)

If your gateway requires an explicit routing config (the `x-portkey-config` header), the recommended approach is to configure a **default config** in the AIGW control plane for your workspace — the gateway then applies it to all requests that don't carry the header explicitly.

Alternatively, you can set it per-project in `.claude/settings.json` at the repo level if Claude Code CLI adds support for custom request headers in your version.

12. Verify with curl

Before starting Claude Code, confirm the token and gateway are working end-to-end:

```

TOKEN=$(./get-az-token.sh) && \
curl http://127.0.0.1:8787/v1/messages \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"model": "anthropic.claude-sonnet-5", "max_tokens": 250, "messages": [{"role": "user", "content": "hi"}]}'

```

Model prefix: when routing via Vertex AI, model names must be prefixed with `anthropic.` (e.g. `anthropic.claude-sonnet-5`). Without the prefix the gateway returns `messages is not supported by vertex-ai`.

If the request succeeds, inspect the token claims to verify the gateway claims are correct:

```

echo $TOKEN | cut -d. -f2 | base64 -d 2>/dev/null | python3 -m json.tool

```

Check that `portkey_oid`, `portkey_workspace` (if used), and `scopes` are present and match what you configured in step 5. For reference, here's a decoded token showing these claims, alongside a second user in a different group picking up different values automatically:

```

{
  "typ": "JWT",
  "alg": "RS256",
  "kid": "dgvSDvK8A5KyKypw1dAwTXD3CA"
}.{
  "aud": "c5490456-8e83-4df6-a979-9ed2b4dd84a5",
  "iss": "https://login.microsoftonline.com/934a04e7-6fbf-4142-83bc-4cecbacabbfb/v2.0",
  "iat": 1790933662,
  "nbf": 1790933662,
  "exp": 1790938281,
  "aio":
"AaQAW/8eAAAAgotN7qR7TX3JqtzD9Fy95jPx8esfvRjGS0Z0vD3goN8CajYH4uLMva001mxKNmumKAvcmQmAKQLu690f2GmkhB6rAsFI620zJW35x19nNPbcy
tyZ/dgW/Yk2P9goNqGQLQ9a5zmgnY0p5VyjUmVEyGE9oQlh2XbgftM7yyewaFYgQU+qGeqqgLzC/GZrI4dnPbAJJrbZj4JJdbPfPg2aLA==",
  "azp": "04b07795-8ddb-461a-bbee-02f9e1bf7b46",
  "azpacr": "0",
  "name": "Marc Gollop",
  "oid": "b473cc00-77d1-4b47-8b66-f5e94f786e11",
  "preferred_username": "mgollop@sase-au.io",
  "rh": "1.AUoA5wRkK79vQkGDvEzsusq7-1YEScWDjvZNqXme0rTdhKUAABxKAA.",
  "scp": "user_impersonation",
  "sid": "008df00a-a519-6980-2f87-11414b2be80c",
  "sub": "xw_xuxN6sDvBlaXjjwVIYs74rWtTLbMf9RPe7o13-R0",
  "tid": "934a04e7-6fbf-4142-83bc-4cecbacabbfb",
  "uti": "uH7GNP2v40yLtruHHl0AA",
  "ver": "2.0",
  "xms_ftd": "u4Dd4k_JiHeDBsIJeoTwn2Sb67nnkoeVa0cKBPcpJbQBamFwYW5LYXN0LWRzbXM",
  "portkey_oid": "95cb67bf-6fb5-4581-ac72-db32e2bb7f2c",
  "portkey_workspace": "ws-main-a-997260",
  "scopes": "completions.write",
  "metadata": "{
    \"environment\": \"production\",
    \"team\": \"frontend\"
  }"
}.[Signature]

{
  "typ": "JWT",
  "alg": "RS256",
  "kid": "dgvSDvK8A5KyKypw1dAwTXD3CA"
}.{
  "aud": "c5490456-8e83-4df6-a979-9ed2b4dd84a5",
  "iss": "https://login.microsoftonline.com/934a04e7-6fbf-4142-83bc-4cecbacabbfb/v2.0",
  "iat": 1790933925,
  "nbf": 1790933925,
  "exp": 1790939576,
  "aio":
"AVQAQ/8eAAAAUsoWxhkSuprACsNybu4m3tk7LVCcNYXWfDYcS+WvJzwcW4WqTzjdWB6iFTM2D9TfSkT+k371tVadsFfUH0oI8jtivZNDtE5FjSS7SndbBI="
,
  "azp": "04b07795-8ddb-461a-bbee-02f9e1bf7b46",
  "azpacr": "0",
  "name": "AIRS Demo User",
  "oid": "c604bcbe-43cb-475a-ae1e-e4ea4ae88b23",
  "preferred_username": "airsdemo@sase-au.io",
  "rh": "1.AUoA5wRkK79vQkGDvEzsusq7-1YEScWDjvZNqXme0rTdhKUAALxKAA.",
  "scp": "user_impersonation",
  "sid": "0098ec8a-ccb3-c990-a152-fe8883135204",
  "sub": "dNznM-MXrQo7wFnwvBk0EPtCisM7A4GgV7gTk4vfsEk",
  "tid": "934a04e7-6fbf-4142-83bc-4cecbacabbfb",
  "uti": "zxHH55QJzUKXaM2fMwxKAA",
  "ver": "2.0",
  "xms_ftd": "iY6Dv_sUx5yRxK_2mQj0g2KbNdH0tcVIX-1ynU3yuzMBamFwYW5LYXN0LWRzbXM",
  "portkey_oid": "95cb67bf-6fb5-4581-ac72-db32e2bb7f2c",
  "portkey_workspace": "ws-main-a-997260",
  "scopes": "completions.write",
  "metadata": "{
    \"environment\": \"production\",
    \"team\": \"backend\"
  }"
}.[Signature]

```

13. Start Claude Code

```
claude
```

On the first run (or after your Azure CLI session expires) a browser window opens to sign in to EntraID. After authentication the session is cached by the Azure CLI — subsequent invocations are silent until the token expires.

To confirm requests are landing in the gateway, check the control plane logs or AIGW analytics for your workspace.

Onboarding a new user

Add them to the Entra group assigned to the app's Enterprise Application (step 4b). No change is needed here — their next `get-az-token.sh` run picks up membership, access, and the correct claims automatically.

Troubleshooting

Symptom	Likely cause	Fix
AADSTS650057: Invalid resource	App not exposed as an API	Complete step 6 — "Expose an API"
AADSTS65001: consent required	Azure CLI not pre-authorized on the resource app	Complete step 7 — "Pre-authorize the Azure CLI"
AADSTS501461: AcceptMappedClaims...	Resource was requested as <code>api://<id></code> instead of the bare GUID	<code>ENTRAID_RESOURCE_URI</code> must be the bare client GUID, not the <code>api://</code> URI — check the <code>env</code> block in <code>settings.json</code>
Token has no <code>portkey_workspace</code> / wrong workspace	User isn't in a group matched by a claim condition	Add the user to the right group (step 4b / step 5)
User can't sign in at all	Assignment required? is Yes but the user isn't in an assigned group	Assign their group under Enterprise Application → Users and groups (step 4)
Claims don't appear in the token	Claims added under the App Registration's Token configuration tab instead of the Enterprise Application	Configure claims under Enterprise Applications → your app → Single sign-on → Attributes & Claims (step 5) — only this system supports the group-conditioned values used here
<code>ENTRAID_CLIENT_ID</code> not set	Variable not in the <code>env</code> block	Add it to the <code>env</code> block in <code>settings.json</code>

Symptom	Likely cause	Fix
<code>messages is not supported by vertex-ai</code>	Model name missing <code>anthropic.</code> prefix	Use <code>anthropic.claude-sonnet-5</code> not <code>claude-sonnet-5</code> when targeting Vertex AI
API errors after copying entra config to global <code>settings.json</code>	Model env vars absent — Claude Code defaults to bare model names without the <code>anthropic.</code> prefix	Add <code>ANTHROPIC_DEFAULT_SONNET_MODEL</code> , <code>ANTHROPIC_DEFAULT_OPUS_MODEL</code> , and <code>ANTHROPIC_DEFAULT_HAIKU_MODEL</code> with the <code>anthropic.</code> prefix to the <code>env</code> block
<code>401 Unauthorized</code> from gateway	JWT invalid or missing claims	Decode the token (see below) and verify <code>portkey_oid</code> and <code>scopes</code> are present
Invalid API Key (Error Code: 03)	Bearer token is a literal string, not an actual token	Use <code>\$(./get-az-token.sh)</code> not <code>{./get-az-token.sh}</code> for command substitution
Browser opens on every run	Azure CLI token cache expired	Normal — sign in; the cache is then reused for the token lifetime (~1 h)
<code>az: command not found</code>	Azure CLI not installed	<code>brew install azure-cli</code>
Gateway returns <code>502</code> / no route	No Portkey config header and no default config set	Set a default config in the AIGW control plane for your workspace

Decode the token to inspect claims:

```
echo $TOKEN | cut -d. -f2 | base64 -d 2>/dev/null | python3 -m json.tool
```

Updating credentials

If you rotate the EntraID app registration or move to a different tenant, update `ENTRAID_CLIENT_ID` / `ENTRAID_TENANT_ID` in:

1. `~/.claude/settings.json` (`env` block) — or the project-level `.claude/settings.json` if you used that approach
2. Re-run `az login --tenant <new-tenant-id>` to refresh the CLI session